Engineering Chemistry · Unit_01 · Molecular spectroscopy, phase equilibria, and computational chemistry

DRAFT — UE26CY151A · Unit_01 · Molecular spectroscopy, phase equilibria, and computational chemistry · Vibrational spectroscopy(anharmonic oscillator)

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

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M1

Vibrational spectroscopy(anharmonic oscillator)

1 Vibrational spectroscopy(anharmonic oscillator)

Study focus: Vibrational spectroscopy(anharmonic oscillator). This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000090 · EV0000091

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We continue with module one on molecular spectroscopy, and in today's class we will look at the anharmonic oscillator model for vibrational spectroscopy.

Key details

- That's because we considered E is equal to half k x squared that is for a harmonic oscillator and that would result in the energy profile being that of a parabola.
- Like we can extend the bond length as much as we want, but still the molecule remains intact is what we see in this curve.
- But the real picture is something different.
- We know that if we stretch a molecule beyond a point, it is going to dissociate.
- So therefore we need to change the model for real molecules from simple harmonic oscillator to that of the anharmonic oscillator.
- As we come closer, we are reducing the bond length because.
- What is the bond length?
- This is the bond length.
- That is the equilibrium bond length.
- But when we come to this point where we have reached the equilibrium bond length of the molecule, then this is the most stable and therefore represented by minimum in the curve, minimum in the energy curve.
- Atoms even further closer, then we are compressing the molecule, and again the energy is going to increase.
- model using the anharmonic oscillator model.
- It says potential energy is equal to d equilibrium 1 minus exponential a times r naught minus r the whole square.

- Till there is zero, and you can very well make out that this would represent the dissociation energy of the molecule.

Example or application from the lecture

- Then we consider and derive the expression for vibrational energy levels for an anharmonic oscillator, and finally we will try to get the vibrational spectrum of an anharmonic oscillator.
- The reason being that when you consider it as a harmonic oscillator, we are assuming that we can go on stretching the molecule and it will remain intact.
- When we say the molecule behaves unharmonically or vibrates in an anharmonic fashion, then the best way to represent the energy profile is given by this curve which is shown in green.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000090 · EV0000091 · EV0000092 · +1 source units in pack.json

Core explanation · 05:00–09:59

And how do we get the energy level expression?

Key details

- So nu E bar because we are representing in terms of centimeter inverse.
- And what is V here?
- V is the vibrational quantum number.
- V is the vibrational quantum number which is going to take up the value 0, 1, 2, 3.
- And therefore by substituting these values of V, V can get the different energy levels which are present in the anharmonic oscillator.
- But only thing here we have some new term here.
- What is this new term?
- and whose value for bond stretching vibrations is always small and positive.
- We have to remember this.
- It is small and positive.
- But so the oscillation frequency expressed in wave number.
- What do these energy levels look like?
- All we have to do is substitute values for V is equal to 0, 1, 2, 3.
- And you can notice here that we have used the anharmonic oscillator model.
- To zero, then we are going to get an energy level here.
- When we substitute V is equal to one, obviously we got this energy level here.
- Again slightly lower than that of the harmonic oscillator.
- When we substitute V is equal to 2, there is again a decrease in energy because of the second term, but remember that this term is getting squared.
- So that means each time we are subtracting a larger number what does it mean V is equal to 2 level will be much more closer to V is equal to 1 as compared to in the harmonic oscillator model and similarly 3 4 5 6 as we go on substituting the.
- Energy levels are going to come closer and closer and we see that as we go to higher levels the vibrational energy levels are actually coming closer.

Example or application from the lecture

- Just like we did in rotational spectroscopy or when we were talking of harmonic oscillator, we have to solve the Schrodinger equation for the harmonic oscillator and the vibrational energy levels now will be given by this expression E that is we are now talking in centimeter inverse so that means the energy in joules divided by H.
- V plus half nu bar was the expression for a harmonic oscillator but here we have a second term because of the anharmonicity and that is given by minus of V plus half squared nu e bar X E.

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Core explanation · 10:00–14:59

What did we see in the case of rotational energy levels?

Key details

- That the energy levels are going farther and farther apart.
- When we came to harmonic oscillator, we saw that the energy gap remains constant.
- You go to higher levels of V also, energy gap remains constant.
- That's why we saw only one line in the spectrum.
- But here, what do we observe?
- Because of this second term getting subtraction.
- And larger and larger term is getting subtracted each time, we find that the energy levels are coming closer together.
- Here we can see that D equilibrium, that is the dissociation energy.
- And there's one more D naught over here.
- What would this be?
- This is the dissociation energy required by the molecule.
- What do we mean?
- Because we know that in the case of vibration,.
- A molecule is always going to possess the zero point energy.
- When a molecule vibrates there should be a change in the dipole moment.
- And what is the specific selection rule now?
- Delta V is equal to plus minus one, plus minus two, plus minus three, so on.
- All transitions are going to be allowed.
- In the case of harmonic oscillator, what was the selection rule?
- Only plus minus one.
- And all other transitions were forbidden.
- But here we have plus minus one, plus minus two, plus minus three, meaning that all the transitions are going to be permitted.
- E versus R.
- And now your energy profile is like this.
- So what it means, delta, and we know that the energy levels are going to be.
- Coming closer like this.

- Just a rough diagram.
- And what do we mean plus minus one is allowed?
- Minus will stand for emission.
- Plus minus one is allowed.
- Then plus minus.
- Two is also allowed.
- And again plus minus three is also allowed.
- So that means you will have a whole lot of transitions that are possible.
- And you can have 0 to 1, 0 to 2, 0 to 3, you can have 1 to 2, 1 to 3, 1 to 4.
- Two to one two to three, two to four, all these transitions are going to be allowed.
- It is going to be very, very complicated and it'll be difficult for us to interpret the spectrum.
- but we have something coming to help us.
- And what is that?
- Most of the transitions now start from V is equal to zero level.
- Because this is the state which is the most populated state.
- All of the transition, most of the transitions are going to start from V is equal to 0.
- Though the selection rule actually allows it.
- But now the spectrum is going to be much simpler.
- Because all transitions will start from these equal to zero level at room temperature when you're recording the spectrum.
- So first one will happen from V is equal to 0 to 1 and that is known as the fundamental absorption.
- Then second one will be 0 to 2, which is known as the first overtone, and then we have v is equal to 0 to 3, that is known as the second overtone.
- From V is equal to zero.
- So you will see in the spectrum fundamental absorption, first overturn, second overturn.
- You can have other transitions also two is equal to four or five, but then those will be very weak in intensity.
- In fact fundamental absorption will be the one which will be most intense and then we have first overtone and second overtone.
- After that the intensity of the transitions become very weak because more energy will be required.

Example or application from the lecture

- If you remember, let us draw this and see what it looks like.

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Core explanation · 15:00–19:59

Fundamental absorption would correspond to V is equal to 0 to V is equal to 1.

Key details

- To 1 and minus V is equal to 0, the expression we get is this mu E bar 1 minus 2 XC centimeter inverse.
- Remember that we are expressing energies in terms of wave numbers.
- Though I have written energy gap here, what do we mean that?
- You all should substitute for yourself and see whether you are getting this expression.
- Go to the energy expression, substitute V is equal to 2 minus V is equal to 0, and this is the expression you get for the first overtone.
- And similarly, second overtone, you.
- Going to get an expression like this.
- We know that the line in the spectrum you're going to see.
- is based on the energy gap because that is the amount of energy the molecule is going to absorb to go from one level to the next level.
- If it was.
- A harmonic oscillator and then we would have seen the line at nu bar, but now there is a 1 minus 2 XC clearly indicating that the energy gap is now less.
- allowed us a plus two and similarly three mue bar if it allowed us a plus three.
- We will see lines in the spectrum.
- will be the first line because the I told you we represent in centimeter inverse energy is increasing this way and that this will require the minimum energy.
- Nevertheless the intensity of this line is going to be very high.
- So you can see here a thick line over here because most of the molecules intensity always depends on the population of the state.
- And since requires minimum energy, though it is a short line it would give you an impression that this.
- Will be the weakest line, but no here the energy gap.
- It's short only because the energy gap is less.
- Therefore, and therefore it appears as the first line.
- And with low energy, and the second line is your first overtone, which is 0 to 2, and that line is appearing here.
- Have 0 to 3 and you have the turn line after this, it actually the intensity goes on falling down, and but one more observation which you have to make here that because these energy levels are coming closer and closer, we will find that the energy.
- Energy gap also decreases as we go further and further in the spacing between the energy lines is also becoming lesser and lesser.
- If you see the lines converging like this, it means that the molecule is showing anharmonic behavior.
- This is what we see at room temperature.
- V is equal to zero, the V is equal to one level also got populated because of the high temperatures.
- Some molecules gained energy and now they are at V is equal to one level.

Example or application from the lecture

- we are expressing energy in terms of centimeter inverse and that's what I told you in the first class that we use spectroscopic units it is just for convenience okay now let us calculate for

first overtone so.
- Let us observe what happens.

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Core explanation · 20:00–24:59

And light infrared radiation is going to fall on this such molecule, then we will find that there will be some transitions happening from V is equal to one also.

Key details

- Minus 3, we will find lines originating from V is equal to 1 as well.
- Nevertheless, the intensity will still be not so high, it will still be less, and therefore.
- We will find that such lines will appear in the spectrum if we do it at a higher temperature, and the reason being the higher vibration levels also are getting populated, and rightly so they are called as hot bands because you see them at higher temperature.
- When you substitute in the energy expression where it changes from goes from 1 to 2, we get this expression nu e bar 1 minus 4 x centimeter inverse.
- If you remember fundamental absorption was nu E bar 1 minus 2 XE.
- This was for the fundamental absorption.
- Is going to this energy gap is further less.
- So, therefore, where will it appear in the spectrum?
- It will appear to the left of the fundamental absorption in the spectrum.
- In the next class, we are going to study about another type of spectroscopy, which is the electronic spectroscopy.

Example or application from the lecture

- Now, if we say let us take an example, we it is at V is equal to 1 and it is going to V is equal to 2.

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A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Vibrational spectroscopy(anharmonic oscillator) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/006_Vibrational_spectroscopy_anharmonic_oscillator__4fe4d43ab2b66d24.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Vibrational spectroscopy(anharmonic oscillator) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/006_Vibrational_spectroscopy_anharmonic_oscillator__4fe4d43ab2b66d24.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Vibrational spectroscopy(anharmonic oscillator) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/006_Vibrational_spectroscopy_anharmonic_oscillator__4fe4d43ab2b66d24.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
Vibrational spectroscopy(anharmonic oscillator) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/006_Vibrational_spectroscopy_anharmonic_oscillator__4fe4d43ab2b66d24.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.

5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00018	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare...
SRC00019	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00020	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare...
SRC00021	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0061. Sidecar variants compare...

C Revision and retrieval practice